Kanghyun Kim

Ph.D. Candidate

Korea Advanced Institute of Science and Technology (KAIST)

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Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, Korea*

Ph.D. Candidate | Department of Electrical Engineering *Sep 2021 - present*

- Research topic: Spatially-Conditioned Diffusion Models: Scene-Level Multi-Fingered Grasp Generation for Diversity and Inference-Time Selectivity
- Advisor: Dr. Min Jun Kim

Sogang University *Seoul, Korea*

M.S. | Department of Mechanical Engineering *Mar 2016 - Feb 2018*

- Research topics: Wearable Robots, Soft sensor system for human intention recognition
- Thesis: An air-filled pad with elastomeric pillar array designed for a force-sensing insole
- Advisor: Dr. Kyoungchul Kong

Sogang University *Seoul, Korea*

B.S. | Department of Mechanical Engineering *Mar 2009 - Feb 2016*

Work Experience

Samsung Electronics *Suwon, Korea*

Control Engineer | Digital Appliance Business *Mar 2018 - Jul 2021*

- Implemented digital controllers for PMSM inverters.
- Developed machine learning inference algorithms for smart home appliances.

Research Interests

- Dexterous Grasp Generation
- Visuomotor Policy Learning
- Task-and-Motion Planning

Research Experience

Intelligent Robotic Systems Lab @KAIST *Daejeon, Korea*

Advisor: Dr. Min Jun Kim *Sep 2021 - Present*

- **Robot Visuomotor Policy from a Single Global View:**
  Developed a spatially conditioned diffusion policy (SCDP) for precise and robust manipulation from a single global RGB view; accepted at **CoRL 2026**.
- **SE(3)-Diffusion for general pose generation from raw scenes:**
  Developed a manifold diffusion model on SE(3) that generates task-relevant poses directly from raw 3D scenes using local geometry-based score prediction; released as an **arXiv preprint**.
- **Reachability-tree based Task and Motion Planner:**
  Developed a novel task and motion planning (TAMP) algorithm that enables mutual guidance between task and motion planners; presented at **ICRA 2023**.

German Aerospace Center (DLR) - Institute of Robotics and Mechatronics *Weßling, Germany*

Advisor: Dr. Jinoh Lee *Oct 2024–Mar 2025; Jan–Oct 2026*

- **Multi-fingered Grasp Generation via Diffusion Models (1st visit):**
  Conducted research on scene-based diffusion models to generate multi-fingered grasps for the dual-arm Rollin' Justin robot. **Under review at T-RO.**
- **Steerable Multi-fingered Grasping Demonstration at the EU Parliament (2nd visit):**
  Conducted an interactive demo with the humanoid robot Rollin' Justin as part of the euROBIN project. Its steerable multi-fingered grasping capability allowed users to specify target objects and grasp types at inference time.

Washing Machine R&D Lab @Samsung *Suwon, Korea*

Project Highlights *Mar 2018 - Sep 2021*

- **On-line Parameter Estimation for Washing Machines:**
  Developed a model-based parameter estimator, achieving continuous laundry weight sensing for dehydration rate detection.
- **Motor Current Signal Classification for Laundry Detection:**
  Developed a method for fabric type identification through motor current analysis based on machine learning, and implemented the algorithm suited for embedded processors.

Exoskeleton Laboratory @Sogang Univ. *Seoul, Korea*

Advisor: Dr. Kyoungchul Kong *Mar 2016 - Mar 2018*

- **Soft Force-sensing Insole System:**
  Developed an air-filled pad integrated with an elastomeric pillar array for soft sensing, and algorithms for human intention monitoring. Published in the **IEEE Sensors Journal**

Publications

K. Kim, A. Köpken, X. Wu, P. Schmaus, J. Lee, and M. J. Kim, "Spatially-Conditioned Diffusion Models: Scene-Level Multi-Fingered Grasp Generation for Diversity and Inference-Time Selectivity," IEEE Transactions on Robotics (T-RO), **In Review**.

S. Kim*, **K. Kim***, D. Ko, Y. J. Heo, and M. J. Kim, "Spatially Conditioned Diffusion Policy: Learning Precise and Robust Manipulation with a Single RGB Camera," Conference on Robot Learning (CoRL), 2026. **Accepted**. *Equal contribution.

K. Kim, M. J. Kim, "SPLIT: SE(3)-diffusion via Local Geometry-based Score Prediction for 3D Scene-to-Pose-Set Matching Problems," arXiv preprint arXiv:2411.10049 (2024).

K. Kim, D. Park and M. J. Kim, "A Reachability Tree-Based Algorithm for Robot Task and Motion Planning," 2023 IEEE International Conference on Robotics and Automation (ICRA), London, United Kingdom, 2023, pp. 3750-3756

K. Kim, S. Shin and K. Kong, "An Air-Filled Pad With Elastomeric Pillar Array Designed for a Force-Sensing Insole," in IEEE Sensors Journal, vol. 18, no. 10, pp. 3968-3976, 15 May 2018

H. Choi, **K. Kim**, P.G. Jung, B. Na, D.W. Rha, K. Jung, and K. Kong, "Ankle Joint Moment Estimation Based on Smart Shoes" IFAC-PapersOnLine, vol. 50, no. 1, pp. 1366–1371, 2017.

Y. Kim, C. Kwon, H. Moon, **K. Kim**, J. Cho, and K. Kong, "Optimization of semi-active pneumatic actuators for an exoskeleton robot for running," in 2018 15th International Conference on Ubiquitous Robots (UR), 2018.

K. Kim, and K. Kong, "Accuracy Improvement Algorithms for Ground Reaction Force Measurement by an Insole-type Sensor System," International Conference on Control, Automation, and Systems (ICCAS), 2017.

K. Kim, P.G. Jung, and K. Kong, "Design of an Elastomer-Based Soft Force Sensor for Measuring Ground Reaction Force in a Shoe," in Proceedings of the Korean Society of Mechanical Engineers (KSME) Annual Conference, pp. 1508-1513, 2016

Patents

K. Kong, **K. Kim**, et al. "Ground Reaction Force Sensor Module, Shoe having the same, Measuring System of Ground reaction Force of Sole and Measuring Method of Ground reaction Force of Sole." KR Patent (Granted)

K. Kim, et al. "Washing machine and control method thereof (Artificial intelligence laundry fabric detection of Washing machine)." URL: https://patents.google.com/patent/US20220243382A1,US20240084490A1/en, US Patent (Granted)

K. Kim, et al. "Washer and control method thereof (Drum inertia sensing method at high speed for washing machine.)." URL: https://patents.google.com/patent/US20220372687A1/en, US Patent (Granted)

Skills

Keywords: Robot Manipulation | Machine Learning | 3D Vision | Numerical Optimization | Embedded Systems

Programming: Python | C/C++ | Matlab/Simulink | LabVIEW | Julia

Robotics/ML: ROS | ML (Pytorch, JAX) | Vision | Motion/Task Planning | Physics Simulation | Optimization

Design: Solidworks, AutoCAD, KiCad

Control HW: Easy-to-use boards (Arduino/Raspberry Pi/Jetson...) | Microcontrollers (ARM Cortex-M) | DAQ (NI myRIO)